

ALEJANDRO MEDINA, MD

Full-Stack Software Engineer | U.S. Citizen
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SUMMARY

Full-Stack Software Engineer and Doctor of Medicine (MD) specializing in healthcare technology and AI-native clinical tools. Expert at translating complex clinical requirements into production-ready code, with a focus on patient safety, data integrity, and physician efficiency. Proven track record leveraging AI-assisted development to ship secure, scalable healthcare workflows.

SKILLS

Languages & Frameworks: TypeScript, React, Next.js (App Router), Node.js, Python (FastAPI), TailwindCSS

Data & Infrastructure: PostgreSQL, SQLAlchemy, Alembic, Pydantic v2, Zod, RESTful APIs, JWT / RBAC

Engineering Practices: AI-Native Workflow, TanStack Query, React Hook Form, Git/GitHub, Vercel, Render

Clinical Domain: EMR Concepts, HIPAA Compliance, Clinical Decision Support (CDS), Patient Safety

EXPERIENCE

Full-Stack Developer / AI Engineer (Freelance) — Remote

2024 – Present

- Architected secure workflows with Role-Based Access Control (RBAC) for sensitive medical data, ensuring full traceability aligned with HIPAA-compliant design patterns.
- Leveraged LLMs to accelerate engineering velocity, automating complex logic generation and testing scripts to reduce development cycles.
- Developed high-performance RESTful APIs to transform complex JSON clinical payloads into structured data for clinical decision support systems.

Medical Intern, Clinical Safety & Operations — Hospital Aranda de la Parra

2022 – 2023

- Managed patient care and clinical documentation in a high-pressure environment, gaining deep insight into clinician workflow pain points and operational frictions.
- Collaborated with multidisciplinary teams to ensure patient safety protocols and data accuracy, directly informing a code-first approach to medical software architecture.

PROJECTS

MediDash — Full-Stack Clinical Dashboard

[GitHub \(Frontend\)](#) | [GitHub \(Backend\)](#)

Next.js 16 · TypeScript · React 19 · TailwindCSS v4 · TanStack Query · FastAPI · SQLAlchemy · Alembic · PostgreSQL · JWT

- Built a production-deployed clinical dashboard for doctors and nurses, enforcing role-based access control (RBAC) down to individual form fields — doctors manage all patient data; nurses are restricted to vitals only.
- Engineered a real-time drug interaction checker using pairwise set logic across a seeded drug catalog, deduplicating symmetric pairs and surfacing severity-level clinical alerts to prevent dangerous drug combinations.
- Implemented JWT authentication with bcrypt, two-layer input validation (Pydantic constraints + SQLAlchemy CheckConstraints), and server-computed patient risk scores including BMI and Glasgow Coma Scale interpretation.
- Delivered a WHO surgical safety checklist system — doctors create pre-populated 10-step checklists; nurses toggle completion with timestamps and per-user attribution tracked on every item.
- Frontend built with React Hook Form + Zod schema validation, TanStack Query for server state, and a custom proxy.ts route guard handling unauthenticated redirects.

EDUCATION

Doctor of Medicine (MD) — Universidad Autónoma de Guadalajara

2023

Full-Stack Web Development (React, Next.js, Node.js) — Udemy

2025

100 Days of Code: Python Pro Bootcamp — Udemy

2025